

SUSTAINABLE ENTREPRENEURSHIP AND INNOVATION

CHAPTER 3: GETTING TO WIN-WIN: THE BUSINESS CASE FOR SUSTAINABLE ENTERPRISE

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8/15/25

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INTRODUCTION

In the last chapter, we discussed win-win and win-lose conditions and argued that undiscovered or under-exploited win-win conditions represent opportunities for the aspiring entrepreneur or corporate strategist. The central question in thinking about the financial implications of sustainable enterprise is: “do actions that help resolve sustainability challenges simultaneously increase economic returns, competitive advantage and/or shareholder value?”. This question has been stated previously as “Does it Pay to Be Green” to “Do ESG (Environmental, Social, and Governance programs) Create Value.”

The answer to this question is, “it depends.” In other words, the financial viability of any sustainability initiative or practice depends on a host of contextual factors such as 1) the nature and priorities of your customer, 2) regulatory drivers, 3) the economics of a given technology, 4) social, economic, and ecological trends, 5) the preferences of investors, and 6) the broader societal context, just to name a few. Thus, the better way to frame the question is, “In what ways do specific efforts to enhance sustainability contribute to the reduction of risk, lowering of costs, or enhancement of revenues?” and “how can I as an entrepreneur or executive generate financial returns to my sustainability mission and activities?” This chapter focuses on the ways in which sustainability practices can contribute to financial performance.

DRIVERS OF ECONOMIC VALUE

From the pragmatic perspective of the executive or entrepreneur making decisions on sustainability initiatives, it pays to understand the ways in which sustainable practices contribute to organizational performance. Or, for sustainability directors or advocates, how do you make the case for sustainability practices to upper management concerned about the financial implications? To understand the business case for sustainable enterprise, the remainder of this chapter discusses five drivers of economic value associated with sustainable enterprise: 1) Cost Reduction, 2) Revenue Generation, 3) Risk Reduction, 4) Employee Recruitment, Retention and Productivity, and 5) Emerging Opportunities (See Table 3.1). Large firms often focus or start with cost reduction, considering it the easiest to justify and the “*lowest hanging fruit*.” By contrast, the most forward-thinking firms focus on emerging opportunities in the global transition to a more sustainable economy. We start with a discussion of cost reduction.

Table 3.1: Drivers of Economic Value from Sustainability Initiatives and Practices

Driver of Value	Actions and Focus
Cost Reduction	Increase efficiency, embrace circular production models, and eliminate materials, energy, and water waste
Revenue Generation	Address changing customer preferences for sustainable B2C and B2B products and services, enhance corporate and/or brand reputation
Risk Reduction	Reduce stakeholder risks: regulatory, reputational, ratings. Reduce Value Chain Risks: physical, price, product.
Employee Relationships	Enhance employee recruitment, retention, and productivity
Emerging Opportunities	Capture opportunities in innovative sustainable business models or emerging sectors such as renewable energy, energy storage, carbon capture, green building, sustainable agriculture, energy efficiency, sustainable transportation

Cost Reduction

One of the primary benefits of pursuing sustainability in organizations is the reduction of costs. Every dollar saved in costs can translate directly to profit, whereas an extra dollar earned in revenue only translates to profit according to how much profit is earned in a dollar of revenue. So, cost savings are an important part of making the business case for sustainability. Fortunately, there are a multitude of opportunities to reduce costs through sustainability initiatives. One of the first companies to comprehensively address environmental aspects of their operations was 3M. In 1975, they started their Pollution Prevention Pays (3Ps) program to enhance their environmental performance. Since that time, they have reduced 2.89 million tons of pollutionⁱ, and saved over \$2.2 billion in costsⁱⁱ.

Waste Not, Want Not. A simple way to look at this is the relationship between waste and cost. Basically, any material leaving an organization as waste was something paid for as an input, and therefore contributes to overall organizational cost. Reducing waste or enhancing efficiency can therefore translate directly into the bottom line. Moreover, something always must be done with waste, and treatment or disposal can be expensive. Waste sent to landfill is subject to transportation costs as well as tipping fees, not to mention the negative externalities associated with landfilling a corporation's solid waste. Other wastes, like water, often need to be treated prior to discharge which can also be costly, whether discharged to waterways, sewage treatment plants, or treated onsite. But the heaviest cost burden often comes from the disposal of hazardous wastes, which typically fall under strict regulatory controls requiring tracking, certification, and ultimate disposal through practices like incineration. Because of this, many companies are highly motivated to avoid creation of hazardous materials in their operations. In any case, waste reduction can substantially lower costs of operations and contribute to competitive advantage. In addition, it is wise to focus on minimizing waste in the entire supply

chain including waste created during the manufacture of inputs, and that generated during and after the use of a product. Life cycle analysis, which we will discuss in more detail later, is a technique for estimating environmental aspects and impacts across the supply chain and therefore an excellent foundation for understanding where waste and impact occur in the entire sourcing, production, use, and disposal cycle. We now turn to brief discussions on various types of waste and how one can reduce costs by minimizing it and increasing efficiency. We start with materials waste.

Materials Waste. Materials are often the first on companies' list and can be the easiest to identify and manage. As noted above, materials waste also translates directly into costs, because it is an input that was paid for but does not provide customer value. Materials waste includes that generated from production and other operations. But a circular economy and life cycle approach suggests that companies should also consider waste that results from the production of raw materials, waste that accrues during a product's use by customers, and disposal after the useful lifespan of a product.

Traditionally, companies employ engineering and accounting methods to estimate the magnitude of materials waste across their operations. These methods involve estimating the differences between inputs and outputs and understanding the causes of differences. Gauging waste can also involve metrics on the quantity of different types of waste leaving company facilities, or even the practice of *dumpster diving* wherein individuals literally sort through company dumpsters to discover exactly what is being disposed. In either case, such metrics and estimations can provide the foundation for waste reduction programs and allow companies to identify priorities.

In setting priorities, it helps to think about the reduce, reuse, recycle approach wherein reducing and reusing materials are prioritized. Reducing can involve dematerialization in products or packaging. In agriculture, smart tractors embedded with GPS navigation and satellite imagery are enabling a revolution in optimizing the application of fertilizers, herbicides, and pesticides, driving down input costs while maximizing productivity and lessening environmental impact. Companies have often had great success in dematerialization of products. Conceptually, one can consider the modern smartphone, which has eliminated the need for a multitude of separate devices including a camera, gps receiver, audio recorder, compass, and entertainment device. In packaging, sometime less is more, and packaging innovation can both reduce costs and enhance customer satisfaction. Hello Fresh, whose business model relies on substantial packaging to ensure safety in their food package delivery service, varies their food packaging according to the location and weather. By minimizing box sizes and ice packs where appropriate, they reduce materials waste, environmental impact, and costs.

Opportunities for reuse of materials can also abound within a company's operations. Materials waste can often be put back into the production process. Some companies even have product take-back programs in which customers return spent products. HP, for example, has recycled over one billion ink and toner printer cartridges since starting their HP Planet Partners program. Customers can take the cartridges to select retailers or ship them directly to HP using return shipping labels included with new toner cartridgesⁱⁱⁱ. HP recycles the plastics and re-uses some parts in new cartridges. In some countries, *extended producer responsibility* regulations force companies to take responsibility for products after use. In other words, companies are required to take back end-of-life products or secure an avenue for recycling or disposal after use. Many argue this increases companies' motivations to design products for durability, remanufacture, disassembly and/or recycling. Other companies, like Xerox, have "servicized"

their business, moving from selling copiers (selling a product) to leasing copiers (a subscription-type service). This approach can secure longer-term customer-relations as well as drive longer product life spans maintained through redesign, repair, restoration, or remanufacturing.

When materials reduction or reuse are not possible, companies can look to recycling programs, which involve making new products or materials from waste. *Upcycling* refers to transforming waste into higher value products than the original materials (for example manufacturing furniture or carpet from recycled plastic bottles). *Downcycling* refers to transforming waste into lower value products (for example, burning plastic bottles to produce heat). Some high value materials like aluminum, copper, or steel can be readily sold. Unfortunately, not all waste can be recycled economically, creating added costs to companies but offering an opportunity for innovation and entrepreneurial action. In these scenarios, companies look for other benefits to incent recycling (such as organizational values, regulations, or fulfilling customer or employee expectations) while looking for creative uses or other solutions.

Companies can look to exemplar organizations when considering innovative ways to reduce waste. For example, New Belgium Brewing in Ft Collins, CO is proud of their anaerobic digester which processes liquid brewing waste and captures the resultant methane for use in their facility. Anaerobic digestion of waste enables the capture of methane for the production of heat or electricity, or a transportation fuel^{iv}. Dairy operations use similar anaerobic digestion technology to reduce greenhouse gases (GHG's) from manure waste. Cattle and sheep production accounts for a substantial percentage of world anthropogenic GHG emissions, most of which come from the methane released from digestion and manure degradation. Large operations managing manure as a liquid use anaerobic digestion to substantially reduce methane release and is being adopted by US cattle operations^v. This appears to be an increasingly popular method, and is being driven by large corporate buyers, potential revenues or savings from captured methane, as well as regulatory requirements in some areas. Cattle operations are also beginning to adopt feed supplements to reduce digestive emissions of methane. One particularly interesting development is the addition of small quantities of bromoform algae (seaweed) to feed to reduce enteric methane emissions by 55%-80% in dairy cows^{vi}.

Many event and concert venues have worked to limit their environmental footprint by transitioning away from plastic beverage cups. Some have adopted cups made from polylactic acid, a polymer made from corn. Marketed by Eco-Products, Inc. the cups are commercially biodegradable and have also been widely adopted by universities focused on reducing their own impact. More recently, many venues have gone a step further and utilize coated aluminum cups made by Ball Corporation. These cups are easily recyclable and have more value to recyclers. While the biodegradable cups are downcycled into compost, the aluminum cups are turned back into aluminum for further use, helping to close the loop.

Energy Efficiency. Like the other types of waste, wasted energy is costly. Moreover, energy derived from fossil fuels has a carbon and climate impact, which is often the top environmental concern for companies. As a result, increasing energy efficiency can have a double benefit – it both decreases costs and a company's carbon footprint. Moreover, there is a general trend of increasing costs of energy as fossil fuel prices rise and the transition to cleaner power requires substantive investment in renewables, storage, and transmission. The International Energy Agency recently reported that energy efficiency investments are at their highest level ever^{vii} and McKinsey analysis suggests companies can save 30-55% in energy costs through strategic investments^{viii}. Because the financial returns can be substantial, enhancing energy efficiency is often seen as the lowest hanging fruit in sustainable enterprise.

Energy comes in many forms. Companies use electricity (power) for heating, air conditioning, manufacturing, and operations, and, increasingly, transportation. Electricity is generated from hydro and non-hydro renewables (e.g. wind, solar, geothermal), coal, natural gas, and nuclear. Renewables and nuclear have a much smaller greenhouse gas footprint and are leading the way in the clean energy transition. Companies also use fossil fuels directly for heating and industrial processes, particularly natural gas. They also use gasoline and diesel for transportation and the operation of machinery such as tractors and construction equipment. Companies can expect increasing costs of energy, particularly with the global trend to internalize the cost of carbon through cap-and-trade regimes, carbon taxes, or other means that make fossil-based energy more expensive. Given the high costs of energy and the climate impact of fossil fuels, attention to energy use is a critical aspect of sustainability efforts in corporations.

Amory Lovins, founder of the Rocky Mountain Institute is famous for emphasizing that the cheapest form of energy is the “negawatt”, or the energy you didn’t have to use to achieve the same result^{ix}. In other words, companies should focus on energy efficiency in their operations to reduce costs. In a 2010 report, McKinsey presented the Global Greenhouse Gas Abatement Cost Curve wherein they identified the least and most expensive climate mitigation strategies^x. The report, though now somewhat dated, laid out the most economically promising actions to reduce greenhouse gases. At the top of the list were numerous technologies that had a net benefit (would save more money than they cost, subject to some assumptions). Most of these were energy efficiency actions like efficient motor systems, advanced lighting, hybrid vehicles, building insulation, heating and air conditioning systems, appliances, and other industry efficiency improvements. In short, McKinsey’s research agreed with Amory Lovins that efforts to enhance energy efficiency have the greatest economic payback. As an entrepreneur or executive this research can be instructive – one of the first steps in making your company more sustainable at the least cost and the greatest return typically involves a focus on energy use and efficiency.

Another aspect of energy use is the increasing drive to price power according to supply and demand conditions, commonly known as *peak pricing*. Depending on the climate conditions and season, power use often spikes in the mid-afternoons. This is known as peak demand and the generation of electricity during peak hours is typically the costliest to produce since equipment must be put in place to meet demand only for a short period of time. The increasing levels of solar on the electric grid are helping when solar production corresponds to hot weather and air conditioning use. On the other hand, it is exacerbating the problem when demand is high, the weather is hot, and the sun starts to go down. One result is the increasing need for energy storage (which also raises costs), but the other is peak pricing, in which utilities raise the price of electricity during times of peak demand. Increasingly, companies will face peak pricing of electricity, which will enhance the returns to energy management efforts.

Water Waste. While the impacts of decreasing freshwater availability may not be as obvious climate change, water scarcity and rising prices will have an increasing impact on companies. Water resources face both decreasing supply and growing global demand resulting in decreasing availability in many areas. Like food scarcity, water scarcity often drives geopolitics, and history suggests can lead to all-out war. Countries battle for water supplies and, even within countries, water access can be contentious. In the U.S., several states are increasingly at odds over the use of water from the Colorado River Basin. While water in some areas is abundant, in others it is scarce and increasing costs on companies are inevitable. Moreover, water quality is often an important aspect of water supply. Beverage companies for

example, often rely on large quantities of high-quality water for their inputs. As a result, firms will have to pay increasing attention to reducing water use. In many areas water supplies will be an obstacle to growth, both for companies and for the growth of populations. Firms will likely find it increasingly beneficial to conserve water and treat water waste for re-use. Firms with high water use will also have to consider water in their locational decisions, with likely migration to areas with sufficient and less costly water supplies.

Revenue Generation

Broadly, enhanced revenue generation from sustainability practices accrues to the ability to differentiate your company or its product from a competitor or its offerings. The goal in such efforts is to differentiate based on sustainability practices and performance directed to market segments that value those attributes. When successful, such differentiation can lead to opportunities to both increase market share or raise prices to enhance profitability. Fortunately, many customers value sustainability attributes. A recent study by McKinsey and Company and Nielson IQ analyzed five years of sales for 600,000 consumer packaged goods and found that those with environmental, social, and governance (ESG) claims grew by 28% compared to 20% for those without such claims^{xi}. Indeed, many consumers respond to social and environmental attributes of the products they buy and the companies from which they buy those products.

Numerous companies, particularly entrepreneurial ones have figured out how to attract markets and build their businesses through sustainability. For some market segments, the benefits have been substantial. From agriculture to construction to renewable energy, companies have been benefiting from the increasing market demand for more sustainable products. In agriculture, the market response to organic foods over the past decades has been nothing short of astounding. In construction, green building, propelled by the LEED Green Building Standards, has grown at an incredible pace. In energy, both residential and commercial customers have voluntarily adopted renewables, helping drive more installation of solar and wind power. In other industries, companies with claims of sustainability often outpace those without such claims. This is especially true for companies serving customers with a sustainability orientation, such those marketing outdoor products and apparel.

But what is often overlooked are the commercial and industrial buyers (B2B) who increasingly look to “green” their supply chains. Often, large companies not only prefer sustainability-oriented organizations that can provide metrics on their activities and impacts but will not buy from companies who don’t manage their social and environmental impacts. Wal-Mart, for example, has set a goal to avoid a gigaton of carbon emissions by 2030 by working with suppliers to reduce their impact^{xii}. Among other initiatives, Wal-Mart created THESIS performance assessments that track key performance indicators of suppliers and helps identify materially significant issues and associated improvement opportunities^{xiii}. The result is a set of standards, requirements, and product specifications that suppliers must meet to be part of the Wal-Mart supply chain^{xiv}.

Large companies, led by leading technology firms, are also getting more selective in their energy purchasing behavior, increasingly buying renewable energy from projects through power purchase agreements (PPAs). Bloomberg New Energy Finance reported that since 2008 corporations have announced solar and wind PPAs of 198 gigawatts of power, more than the generation capacity of countries like France and the United Kingdom^{xv}. Among the top purchasers of renewable energy PPAs were Amazon, Meta, Google, and Microsoft^{xvi}. Companies purchase renewables for a variety of reasons, but at the top are clean energy goals that contribute to their reputation, and the improving economics of clean power (Same citation). Regardless, such sustainability goals create opportunities for providers of renewable energy or other sustainable products and services who can help large companies meet their sustainability aspirations.

Companies large and small also seek to increase revenue through the positive impact of sustainability practices on their overall brand or reputation. For small companies, sustainability strategies often support a brand proposition based on authenticity. For larger companies, sustainability initiatives can garner customer loyalty and enhance corporate reputations. To the extent that customers affiliate with brands according to their sustainability or social responsibility programs, such programs can impact buyer behavior. Many leaders embrace certification programs like B-Corp that signal their commitment to sustainability.

At the same time, the extent of demand for sustainability-oriented offerings is often overstated, as the drive for lower priced goods often makes it difficult to compete with sustainability attributes that add costs beyond their benefit in product differentiation. As such, entrepreneurs and executives need to think deeply about the extent to which sustainability attributes impact demand and pricing. While such considerations are somewhat nuanced, they are also increasingly ripe with opportunity.

Employee Recruitment, Retention, and Productivity

In addition to cost savings and potential revenue generation, focusing on organizational sustainability can create substantive returns with respect to employees. As employees become increasingly selective in their career choices, many migrate toward companies with clear sustainability missions in either their products, or in the way they operate their businesses. In short, sustainability initiatives can be valuable in the recruitment, retention, and productivity of their work forces, which can lead to both organizational efficiency and effectiveness.

In 1997, Steven Hankin of McKinsey & Co. first coined the term the “war for talent.” Hankin described a shift in demographics in OECD countries characterized by simultaneous increasing demand and decreasing supply of human capital. Fast forward to the 2020s, the demographic trends of increasing demand-decreasing supply has only intensified as firms across all sectors of the economy – both in the developed world (OECD countries) and the developing world (BRICS countries) – struggle to recruit and retain talent (Auginis et al. 2012)^{xvii}.

Added to this global trend, is cultural change towards greater work-life balance and demand for employee-employer values alignment. These two conditions are associated with stakeholder capitalism, a system in which firms are increasingly encouraged to serve the interests of all of their stakeholders, not just the interests of its shareholders, and has been championed by the *Millennial* and *Gen-Z* populations^{xviii xix}.

Increasingly, workers of all ages, especially workers from the *Millennial* and *Gen-Z* populations, are demanding that firms implement HR policies that emphasize work-life balance^{xx}. The trend toward work-life balance intensified during the Covid pandemic as workers

across the globe faced additional stressors and constraints. Central in employee demands for greater work-life balance is the availability of flexible work arrangements, including remote work, flex-time, compressed workweeks, and job-sharing opportunities^{xxi}. There are pragmatic arguments for flexible work arrangements including employee satisfaction, gender equity, family priorities and for firms, organizational commitment, higher levels of productivity, and recruiting. There are also some clear sustainability benefits for companies. First, flexible work arrangements ease society's "commute burden", which can produce energy and time efficiencies^{xxixiii}. In addition, flexible work can lead to better physical, mental and emotional health of employees, which reduces "wasted" energy on overtaxed governments and their health service organizations^{xxiv}. Lastly, providing flexible work allows firms to downsize their operational carbon footprint, and scale back investments in energy-intensive real estate^{xxv}. This point is most acute in older cities with an aging stock of buildings (e.g. NYC or London) that require significant investment to make them carbon neutral in the future.

Perhaps even more poignant for many *Millennial* and *Gen-Z* employees is their preference for organizations that align with their values and/or reflect a stakeholder-centric orientation^{xxvi}. This alignment starts with greater awareness for and commitment to ESG in general (e.g. DEI - diversity, equity & inclusion practices), and climate change in particular^{xxvii}. While much momentum developed within the *Baby Boomer* and *Gen-X* cohorts, *Millennial* and *Gen-Z* populations are demanding even greater commitments from firms to align their business practices with the UN's Sustainable Development Goals (SDGs). For example, recent data from the Pew Research Center notes that climate change and sustainability are a more critical issue for *Millennials* and *Gen-Zers* than their *Baby Boomer* and *Gen-X* brethren^{xxviii}. In fact, these employees are more likely to think about and discuss solutions to climate change, and importantly feel more urgency for climate action than *Baby Boomers* and *Gen-Xers*^{xxix}. In the US, these populations are strong supporters of climate, energy and environmental policies, and want their employers to actively support climate-centric programs. Increasingly, firms need to actively engage in "greening" their business activities, and to take steps to transform their businesses to be more sustainable or, if necessary, divest business lines that do not align with the UN's SDGs^{xxx}.

In the global War for Talent, firms need to transform their HR practices and policies for their *Millennial* and *Gen-Z* employees who seek greater work-life balance, a diverse and inclusive workplace, and an employer focused on sustainability issues and climate change. While there exists exemplar firms who have demonstrated better "values alignment" with these populations (e.g. Patagonia), the majority of firms are still at the early stages of transformation to put a more stakeholder-centric orientation into practice.

Regardless, organizational commitments and actions on sustainability can enhance employee relations and help firms recruit and retain talent, as well as enhance organizational productivity.

Risk Reduction

Companies also face substantial risks related to social and environmental issues. These include risks arising from stakeholder responses to sustainability challenges, as well direct risks in their value chains^{xxxi}. Sustainability actions can help to mitigate these risks, and thereby impact a company's bottom line, especially in the long term.

Foremost among *stakeholder risks* are regulatory risks, reputational risks, and ratings risks. Regulatory risks are those that involve the possibility of regulatory interventions that

adversely impact companies. These may occur at the national, regional, or local levels and have often taken the form of command-and-control regulations that force companies to adopt certain abatement technologies, limit emissions or constrain the use of certain inputs. Initiatives that proactively solve sustainability issues can prevent government regulatory action if pursued effectively. Often, however, such actions need to span industry players to be effective, and therefore require collective action or voluntary initiatives that engage multiple industry members. In any case, the proactive management of sustainability issues and regulatory developments is an important aspect of sustainability efforts in corporations. The goal is to help the industry move to more sustainable practices while trying to ensure that regulatory interventions create less financial harm to the firm and the industry.

Reputational risks are also critical, and sustainability practices that prevent reputational damage can be invaluable. At the least, managing sustainability issues can prevent major incidents that damage corporate reputations and lead to customer boycotts or lawsuits. Reputational risks that result from problems in a company's supply chain are among the most difficult to manage. Business history is rife with examples of supply chain problems that damaged corporations, including the Union Carbide Bhopal Incident in which the release of poisonous gas killed thousands, the Nike sweatshop labor revelations in which the company was accused of using child labor and allowing poor working conditions among its suppliers in east Asia, and the collapse of a garment manufacturing facility in Bangladesh in which over 1,000 people perished. More broadly, attention to sustainability issues can focus management attention on areas that leave the reputation of a company and its brands vulnerable to criticism.

Relatedly, all companies operate under what is known as a "social license to operate."^{xxxii} Social license to operate refers to the acceptance and approval of a company and its operations by a local community or broader stakeholders. First popularized in the mining industry, the concept refers to the fact that all companies require the acceptance of society for their existence, and that failing to meet a set of expectations of local stakeholders and broader civil society will preclude a company from continuing operations. In the mining industry, failure to meet such expectations could result in limits to further expansion in a country, or inability to operate in future locations due to behaviors in prior sites. The concept is gaining popularity in other industries, such as forestry, but it reflects the longer-term issue all companies could face if they fail to live up to societal expectations. While such outcomes are unusual, companies who incur extreme reputational hits for bad behaviors can lose their social license to operate and face dire consequences, as did Purdue Pharmaceuticals in the opioid scandals associated with the marketing of OxyContin.^{xxxiii}

Ratings risks are another increasingly important aspect of sustainability. The increasing investor focus on sustainability and ESG (Environment, Social, and Governance) issues has resulted in a host of ratings and metrics that investors use to make decisions on investment allocations. Companies pay increasing attention to such ratings as interest in them has grown, and ending on the wrong side of such ratings can have substantive impacts on stock price. Moreover, many such ratings indicate real financial risks (from, for example, increasing climate change impacts and policy) that effect longer term returns, and associated investment preferences. The bottom line is that such ratings matter and a proactive approach to sustainability can prevent negative ratings that impact investment decisions and access to capital.

Value chain risks include physical risks, price risks, and product risks and are especially prevalent around issues of climate change (Engel et al., 2015, same cite as above). Physical risks include potential damage to facilities and infrastructure that impact business operations,

including those in the supply chain. With increasing weather events such as hurricanes, drought, and floods, facilities are increasingly vulnerable across the globe. Massive storm damage to refineries can impact operations of oil companies but also impact price and supply conditions for critical inputs. Drought can impact agricultural operations and the availability and price of agricultural inputs to consumer packaged-goods companies. Sea level rise can have dramatic impacts on corporate facilities, real estate values, and insurance costs. While it is sometimes difficult to predict such events, companies can certainly benefit by understanding the nature of physical risks, mitigating for potential impacts, and adjusting operational tactics accordingly.

Price risks refer to potential impacts on the price of commodities from climate change or other sources of ecological damage that impact productivity and supply conditions. Commodity prices are often volatile, but climate change can further impact supply conditions and the cost of inputs. Companies can at least analyze the nature of potential changes, attempt to mitigate for the impact of major disruptions, and hedge against unforeseen price volatility.

Product risks are particularly concerning for some firms whose product portfolio is at risk of becoming unpopular, costly, or unsellable due to sustainability issues. Again, climate change is a central driver in such risks, if the costs of carbon emissions rise or social preferences shift away from damaging product portfolios. If, for example, global trends shift away from the consumption of fossil fuels, many producers are at risk of obsolescence in their fossil-fuel reserves and central product lines.

Overall, paying attention to sustainability issues in your company can help mitigate many risks from the global trend toward more sustainable business practices. Companies who see this future can manage and avoid the stakeholder and value chain risks they face as they navigate the future.

Emerging Opportunities for Innovation

Paying attention to sustainability can also be a driver of new innovations and product portfolios that generate a substantive economic return. As we discussed in Chapter 2, as global society adjusts to new conditions, economic institutions will increasingly reward companies for the pursuit of more sustainable innovations, technologies, and business models. As such, companies who are ahead of the game will discover new opportunities and find markets for their sustainable innovations and business models^{xxxiv}. First, engaging with suppliers on sustainability initiatives can reveal new opportunities for efficiency, cost savings, or product design that minimize impacts and increase customer value. Second, life cycle analysis of product lines can also reveal the most harmful products or the most problematic aspects in the life cycle of a product. After realizing that the most ecologically impactful and costly aspect of their laundry detergents was the heating of water, Procter and Gamble reformulated their product and launched Tide Coldwater^{xxxv}. Adoption rates of cold water washing in some countries like Britain and Holland increased substantially, and customers saved on their energy bills while P&G benefited from a new product line and reputational effects. Third, companies can use sustainability considerations and knowledge to develop entirely new business models that reduce environmental impacts. Understanding impacts across the life cycle can lead to new ideas like servicing your product line (as Xerox did with its copier business), or completely altering delivery methods.

But most importantly, understanding emerging sustainability issues can focus companies on the nature of their business portfolios and allow them to realign their portfolios and business ideas with emerging sustainability trends and challenges. The goal is to capture the business

opportunities of the future, as opposed to the one's of the past. The central message of this text is that new innovations in sustainable enterprise are one of the biggest economic opportunities available to entrepreneurial firms. And other authors contend that a primary benefit of pursuing sustainability missions and practices is that it is a central driver of innovation^{xxxvi}. For large companies and executives, this means identifying unique competencies that can serve emerging business models and sectors. For small companies and entrepreneurs, this means identifying unique niches emerging in the transition to sustainable economic systems.

One need only look at developments over the last few decades to realize global opportunities for sustainable enterprise have been increasing rapidly. Accounting for less than 2% of global power generation in the year 2000, non-hydro renewables like solar and wind accounted for 16% of total global electricity production in 2023. Including hydropower, the total is more like 30% of all power generation^{xxxvii}. The growth has been nothing short of astounding in such a short period of time. Non-hydro renewables are mostly large-scale wind and solar photovoltaics, but other technologies like concentrating solar power, tidal power, and geothermal are also playing a part in this growth. Wind and solar power have benefitted tremendously from declining costs associated with learning curves and economies of scale. Large-scale wind power was the first technology to experience substantial declines in cost as blade size increased and efficiencies grew. More recently, however, the cost of solar photovoltaics (pv) declined precipitously and has driven increasing capacity expansion throughout the world. The improving economics of renewables have driven the expansion of these industries and spawned a host of new ventures and corporate expansions. In the west, Vestas wind has grown rapidly. And Chinese startups in solar pv manufacturing have come to dominate the industry. Smaller players focused on residential and commercial installations have burgeoned across the United States and elsewhere, demonstrating opportunities accruing to both large corporations and innovative or first-moving start-ups. Moreover, renewables' share of power production is expected to rise to 65-85% of total power production by 2050^{xxxviii}.

At the same time, the variability of renewables like solar and wind power have created some problems for energy grids, particularly as the percentage of renewables on the grid increase substantially in places like California. This variability increases the need for alternative sources of power production when renewables are not producing. Natural gas power production has played a critical role in balancing power loads, particularly in places where natural gas prices have declined. However, energy storage technologies are playing an increasing role and are expected to grow substantially. With the decline of lithium battery prices, many utilities are installing lithium-ion battery systems that help solve this problem as well as provide ancillary services that help stabilize the electric grid. In addition, several innovative companies have been working on solutions that embrace cutting edge technologies like gravity storage, compressed air storage, flywheel storage, underground pumped storage, and hydrogen storage. All represent potential economic opportunities for innovative entrepreneurs and corporations.

The declining costs of lithium batteries have also helped revolutionize the transportation industry. Spurred by declining costs, policy incentives, increasing demand, and early offerings by Tesla and Nissan, most major automotive manufacturers are now offering multiple models of electric vehicles (EV). Fully electric and plug-in hybrid electric vehicle global sales grew by 57% in 2022 and are expected to account for 18% of the automobile market in 2023^{xxxix}. In 2023 60% of EV sales were in China, while 25% and 10% were in Europe and the United States, respectively^{xl}. Moreover, EV prices are decreasing and the penetration of electric drivetrains in buses and truck is growing. Sales of electric bicycle are also rising rapidly. To the extent they

replace automotive use for trips around town, they can also have substantive impacts on greenhouse gas emissions. Battery manufacturers like Samsung are also working on solid state EV batteries that are safer, lighter, smaller and charge faster. If costs can come down, such technologies have the potential to further revolutionize the market for EV's. The advent of EV's created tremendous economic opportunities for EV manufacturers like Tesla, Rivian, and BYD, and NIO, not to mention the opportunity for the incumbent carmakers, bicycle manufacturers, battery manufacturers, charging system installers, maintenance and repair facilities, and battery recycling facilities. The International Energy Agency argues that if existing national energy and climate targets are met on time, two-thirds of all vehicles sold in 2035 would be electric^{xli}.

Although it is still early, recent efforts to capture the benefits of hydrogen are also promising^{xlii}. Hydrogen is the lightest element and serves as a fuel in the form H₂. Most hydrogen is currently produced from fossil fuels wherein CO₂ is released in its manufacture. Such hydrogen is termed black, brown, or grey hydrogen. Its carbon impact makes it substantially less attractive as a fuel in helping with climate change issues. However, hydrogen can also be produced through electrolysis, a process which uses electricity to split water molecules into hydrogen and water. Combined with electricity from a renewable source, such as solar or wind, hydrolysis results in a largely carbon-free source of hydrogen that is referred to as green hydrogen. Once obtained, green hydrogen can serve as the foundation for a clean-energy economy. First, it can be burned as a fuel to replace fossil fuels for heat in industrial processes. It may also be used as a fuel for transportation including trucking, shipping, or automobiles, although some argue that its use in personal vehicles is impractical for several reasons. It can also be processed in fuel cells which create electricity by combining hydrogen with oxygen, making water in the process. One potential application of green hydrogen fuel cells is in transportation. Another is energy storage. When variable renewable sources (i.e. solar or wind) are generating extra electricity, that power can be diverted to an electrolyser to produce hydrogen, which can then be stored. When additional power is needed on the grid, the hydrogen can be processed in a fuel cell to produce electricity, effectively providing the energy storage solution needed to compensate for the variability of renewable sources of power. Hydrogen technologies and their applications are largely in their formative stages, with some early-mover companies like Bloom Energy, Ballard Power, and Plug Power leading the way. But large corporations have been making massive investments in hydrogen technologies and applications. Major players investing in hydrogen include BP, Air Products, Linde, and a host of technology start-ups funded by venture capital.

The world is also witnessing gains from various methods of sustainable agriculture. As mentioned above, technological developments in traditional agriculture are increasing yields and reducing the application of pesticides and fertilizers. Organic agriculture prohibits the use of synthetic fertilizers and pesticides, genetically modified breeds, and certain additives and preservatives. In the U.S. organic food sales topped \$60 billion in 2022 and accounted for 6% of total food sales. The industry spawned numerous entrepreneurial ventures that have become household names such as Horizon Organic, Silk, Organic Valley, Annie's, and a host of smaller producers that offer meats, eggs, produce, and packaged food products. Recent trends are taking organic a step further and marketing products as regenerative agriculture which focuses on carbon capture, soil health, biodiversity, and resilience^{xliii}. Companies like Mars, Danone, Cargill, and Diageo are working with or experimenting with regenerative agriculture suppliers. After analyzing the risks of extreme weather events on its potato harvest, McCain Foods (which supplies a quarter of the world's French fries) committed to rolling out regenerative agriculture

across all its potato acreage by 2032^{xliv}. Such trends bode well for farmers and others interested in more sustainable approaches to agriculture.

Building practices are also being transformed, creating tremendous opportunities for architects, builders, and suppliers of more sustainable building materials. Sparked by the development of Leadership in Energy and Environmental Design (LEED) a green building certification program in the U.S. and the Building Research Establishment Environmental Assessment Method in the U.K, green building has soared in the last two decades. Promulgated by the U.S. Green Building Council, LEED uses a point-based system to rate commercial, residential buildings and remodels based on health, efficiency, and sustainability. The market size for green buildings was estimated at \$.59 trillion in 2024 and is expected to grow over 10% CAGR to \$1.1 trillion by 2029^{xlv}. The green building industry is also being helped by government policies and incentives that promote more energy efficient and sustainable buildings and constructions.

Finally, it bears mentioning the multitude of technology firms that are often ignored in discussions of sustainable enterprise. They are arguably the unsung heroes of sustainability. Craigslist, Ebay, and now Facebook marketplace have transformed the market for used products. Prior to their development, the transaction costs of selling or finding buyers for used goods was too high to warrant their sale. Twenty to thirty years ago the only options were a garage sale, newspaper classified, donation to a thrift store, or the landfill. Through online platforms such as craigslist, old products are much more likely to gain a second life. Sharing economy firms also contribute to sustainability by reducing the need for additional production. Cars, tools, commercial kitchens, and even homes can now find multiple users, potentially reducing the need for more cars, buildings, or other goods. Zoom, Microsoft Teams, Apple Facetime and other video-conferencing technologies stimulated by the COVID pandemic have drastically reduced the need for commuting in many businesses, as well as reduced the need for office space. These technologies are also making remote medicine more viable, both reducing the time and expense of traveling to appointments and expanding the availability and efficiency of medical services. The take-away from these examples is that the sustainability benefits of some companies are often hidden and unrecognized. The competitive drive for efficiency and effectiveness often translates directly into sustainability benefits, whether or not this is the intention or focus of a company or its founders.

Overall, these growing sectors exemplify an emerging and dynamic set of opportunities for astute entrepreneurs and innovators. Understanding these emerging sectors and technologies and acting to exploit them is perhaps the greatest opportunity for returns and impact.

PREPARING FOR TOMMORROW.

Given the extent of challenges and increasing pressure for companies to lead and manage their impacts, the returns to sustainability initiatives and innovations will only increase with time. Paying attention now and preparing for the future will pay dividends as we increasingly recognize the challenges of climate change, biodiversity, food supply, and more. The business case for sustainable enterprise will only be enhanced by increasing costs of inputs, increasing market demand, regulatory oversight, changing incentives, and the need to attract and retain the most talented employees. Companies and their leaders need to approach their sustainability initiatives thoughtfully and prioritize efforts that enhance their competitive positions both today and for tomorrow.

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